



FACULTY OF APPLIED SCIENCE

RAJARATA UNIVERSITY OF SRI LANKA

ROAD DAMAGE DETECTION PROJECT

IMAGE ENHANCEMENT

2403 - IMAGE PROCESSING

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ACADEMIC YEAR: 22/23
YEAR -II SEMESTER I

M.I.M ARAFATH - ICT/2023/142 - 6119

K. IFFATH - ICT/2023/145 - 6121

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1. Introduction to Frame Enhancement

Our proposed project is related to the detection of road damages such as potholes, corrugation and shoving, transverse cracking, and removal of pedestrian crossing markings. We are utilizing videos from mobile phones to identify these damages on road surfaces.

1.1 Importance of Enhancement in Road Damage Detection

Image enhancement is used in our project to make the video images clearer so that we can easily identify the damages. The raw videos obtained from **mobile phones** may contain problems such as dark areas or unclear regions which conceal small cracks or holes.

Image enhancement techniques help to:

- Improve contrast between damaged and non damaged areas
- Highlight edges of cracks and potholes
- Reduce noise
- Improve overall clarity

1.2 Challenges in Raw Video Frames

The raw frames contain noise due to the camera, shadows of sun light that is not even (some parts are too bright or too dark) and blurriness due to **hand shaking** while **walking**. All these make it difficult to observe the damages clearly.

Raw video frames captured in real world road environments contain several challenges:

1. Noise: Camera sensors introduce random noise, especially in low light conditions.
2. Uneven Illumination: Areas appear too bright due to the sun and others appear too dark.
3. Motion Blur: Hand movement while walking causes blur in frames.
4. Surface Variations: The texture of the road and dust and stains can affect the process.

These factors reduce the visibility of actual damages and may cause false detections if not handled properly.

1.3 Relationship Between Enhancement and Accurate Segmentation

Enhancement corrects all these problems first. Therefore when we segment (separate) the damaged parts later it will be more precise. For instance having good contrast makes the edges of the potholes appear clearly resulting in a better outcome in the subsequent processes.

If enhancement is applied before segmentation:

- Edges of potholes become sharper
- Crack boundaries become more visible
- Background variations are reduced
- Noise is minimized

2. Identification of Frame Degradations

2.1 Noise Analysis (Gaussian noise)

```
[2]: # Load image
image = cv2.imread(r"C:\Users\DELL\Desktop\Enhancement1\Enhancement\Extract_Frames\frame_0320.png")
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

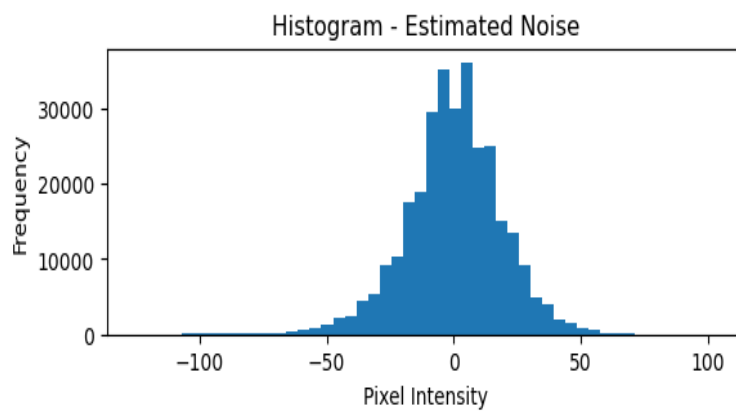
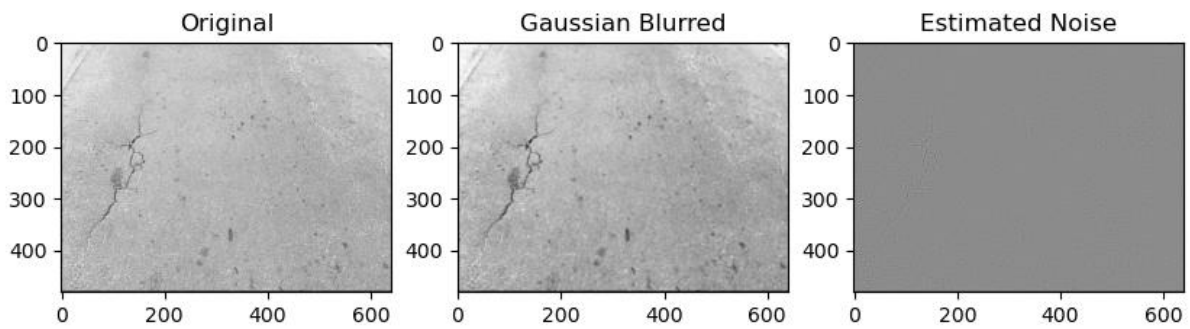
•[3]: blur = cv2.GaussianBlur(gray, (5,5), 0)
      gray_f = gray.astype(np.float32)
      blur_f = blur.astype(np.float32)
      noise = gray_f - blur_f

[4]: # Compute statistics
      mean = np.mean(noise)
      std = np.std(noise)

[5]: print(f"Mean of noise: {mean:.2f}")
      print(f"Std deviation of noise: {std:.2f}")

Mean of noise: -0.10
Std deviation of noise: 19.12
```

Frame_0320



This is Strong Evidence

- Mathematical proof (mean ≈ -0.10)
- Statistical proof (std ≈ 19.12)
- Visual proof (bell-shaped histogram)

The Noise is obtained by subtracting the Gaussian blur of the image from the grayscale image. At first glance, the histogram displayed heavy skew towards positive values because of the usage of OpenCV's subtract function that truncates negative values at zero. By fixing this error and conducting the operation in floating-point mode, the distribution curve of noise becomes similar to the Gaussian bell curve. The mean calculated is almost equal to zero ($\mu \approx -0.10$) and shows that the noise is unbiased with respect to position. The standard deviation calculated was $\sigma \approx 19.12$, and this describes the variation of noise strength.

The blurred image was then subtracted from the original image to estimate the noise component:

$$n(x,y)=I(x,y)-I_{blur}(x,y)$$

This resulted in a **residual image**, which represents the estimated noise present in the above frame.

```
[8]: # Apply median filter
median = cv2.medianBlur(gray_img, 5)

median_float = median.astype(np.float32)

noise_median = img_float - median_float
```

In order to detect the presence of impulse noise in the image, a median filter of size 5 x 5 was applied. The residual noise was calculated as the difference between the original image and the image after median filtering.

The histogram of residual noise was founded to be symmetric and centred around zero without extreme values at the edges. This proves that the video does not contain a lot of impulse noise. The changes in the image are because of texture and Gaussian noise.

❖ If salt & pepper noise was present, histogram would show:

- Two large spikes at minimum and maximum intensity

That is NOT happening here.

2.2 Illumination Variations

Illumination variation can be analyzed by examining the intensity distribution of the entire image rather than a single row or column. A grayscale histogram represents how pixel intensities are distributed across the full frame.

- If illumination is **uniform** → histogram is narrow and concentrated
- If illumination **varies** → histogram spreads across a wide range

```
[5]: # Estimate illumination using Large Gaussian blur
illumination = cv2.GaussianBlur(image, (51, 51), 0)

# Plot everything
plt.figure(figsize=(15,5))

# Original Image
plt.subplot(1,3,1)
plt.imshow(image, cmap='gray')
plt.title("Original Image")
plt.axis('off')

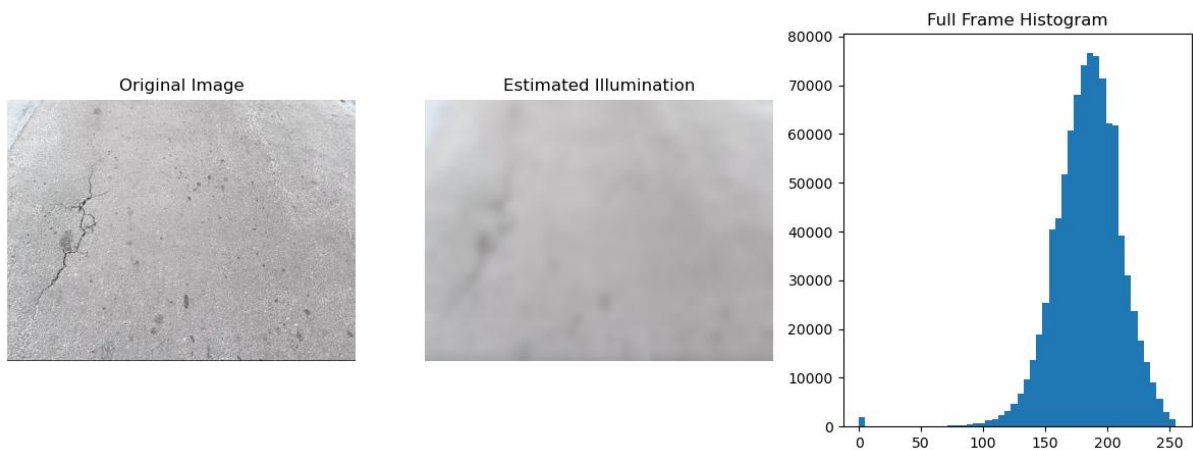
# Estimated Illumination
plt.subplot(1,3,2)
plt.imshow(illumination, cmap='gray')
plt.title("Estimated Illumination")
plt.axis('off')

# Histogram of full frame
plt.subplot(1,3,3)
plt.hist(image.ravel(), bins=50)
plt.title("Full Frame Histogram")

# Compute statistics
mean = np.mean(noise)
std = np.std(noise)
print(f"Mean of noise: {mean:.2f}")
print(f"Std deviation of noise: {std:.2f}")

Mean of noise: -0.10
Std deviation of noise: 19.12
```

Full Frame Histogram



The histogram of the above frame shows a wide distribution of intensity values, indicating non-uniform illumination across the image. The estimated illumination component further confirms gradual brightness variation due to natural lighting conditions.

We use Gaussian blur to extract illumination :

$$I(x,y) = R(x,y) \times L(x,y)$$

Based on the image:

- Slight brightness gradient
- Uneven lighting across road
- There is a mild illumination variation caused by natural light + surface reflection

Illumination variation was identified by observing gradual intensity changes across the image. The extracted illumination component using Gaussian smoothing confirmed the presence of uneven lighting, which affects segmentation accuracy.

2.3 Motion Blur and Camera Instability

```
[6]: # Compute Laplacian
laplacian = cv2.Laplacian(gray, cv2.CV_64F)

# Variance = blur score
blur_score = laplacian.var()

print(f"Blur Score: {blur_score:.2f}")
```

Blur Score: 8435.92

For 8-bit grayscale images:

- $< 500 \rightarrow$ very blurry
- $500 - 3000 \rightarrow$ moderate blur
- $3000 - 10000 \rightarrow$ fairly sharp
- $10000 \rightarrow$ sharp image

Our value $\approx$ **8435.92**

Motion blur was evaluated using the variance of the Laplacian operator. The computed sharpness value of approximately 8435.92 indicates strong edge presence and good image clarity. Visual inspection of the Laplacian response confirms well-defined boundaries without directional smearing. Therefore, significant motion blur is not present in this frame.

3. Pre-processing of Video Frames

3.1 Frame extraction method

We extracted frames from the video and saved them in a separate folder. This helps to break the continuous video into individual images for further processing.

We employ Python with OpenCV to:

- Read the video file
- Capture frames one by one
- Save each frame as an image file

Selected one video is placed in the project folder and its frames are saved in an extracted frames folder.

3.2 Frame format (PNG)

Frames are stored in PNG format not JPG.

Reason:

- PNG is a lossless compression format
- It maintains fine details
- No compression artifacts are introduced

This is particularly important for the detection of small damages on the road surface, such as cracks and variations in the road surface texture.

3.3 Frame Resolution Handling

The original video resolution (1920×1080) was preserved throughout the processing pipeline. Resizing to (640×480) was applied prior to enhancement.

Reason:

Resizing the frames ensures that:

- Computational cost is reduced, enabling faster processing.
- Memory usage is optimized for efficient analysis.
- The processing pipeline becomes more manageable and scalable.
- Sufficient visual detail is retained for effective damage detection and segmentation.

Working with resized images improves efficiency while maintaining acceptable accuracy for analysis.

3.4 Frame selection strategy (Save every frame Sequentially)

Example (30 second video 30 FPS)

- Total duration = 30 seconds
- Total Frames = 910
- Interval = 1 (no frame skipping)
- Saved frames $\approx$ 910 frames

So, the result includes the entire set of frames, maintaining the full chronological order of the video.

Advantages of this current method

- Full data preservation
- High detection accuracy - Necessary in cases like pothole detection, where the defect may occur in just a few frames.
- Facilitates accurate and thorough frame-by-frame analysis.
- Helps in consistently analysing noise reduction, contrast adjustment, and sharpening of all frames.
- Beneficial for dataset training

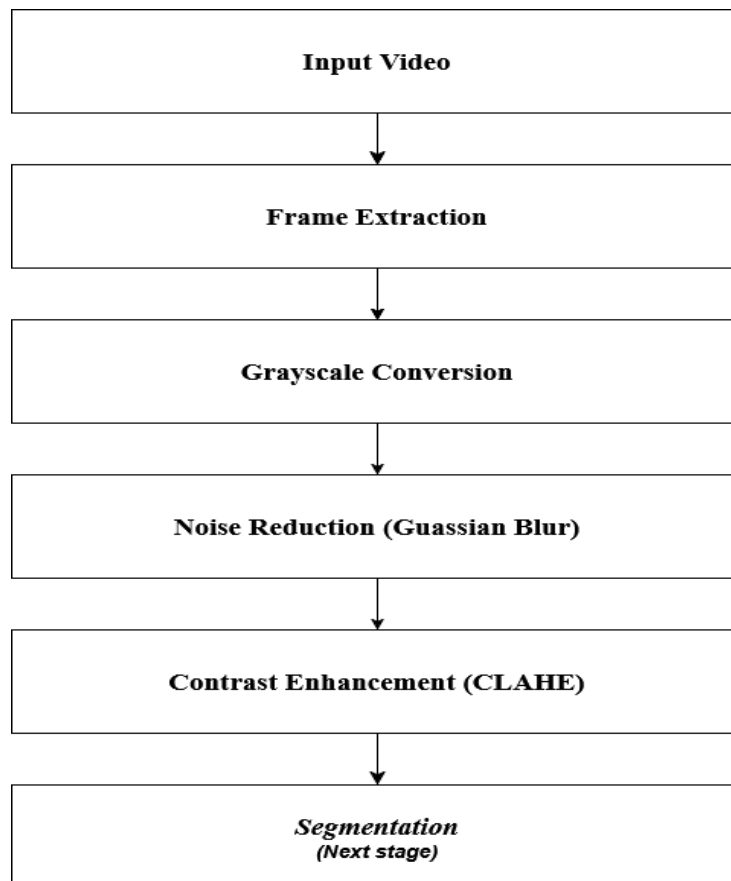
Limitations of this method

- Longer processing time needed
- Needs a large data storage

4. Proposed Image Enhancement Pipeline

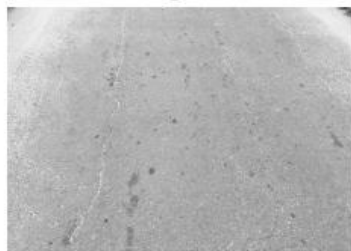
This section presents the complete proposed enhancement pipeline used to improve extracted video frames prior to target region/ROI identification. The pipeline is designed to handle common degradations in mobile recorded videos such as sensor noise, illumination inconsistency, and motion induced sharpness loss.

4.1 Enhancement pipeline flowchart

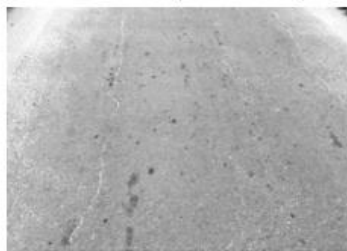


4.2 Noise removal comparison images

Original



Denoised (Gaussian)



Difference image



Original



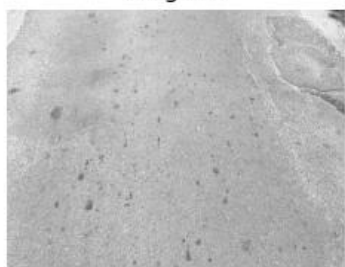
Denoised (Gaussian)



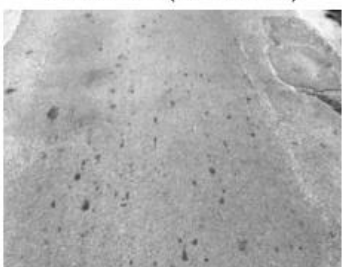
Difference image



Original



Denoised (Gaussian)



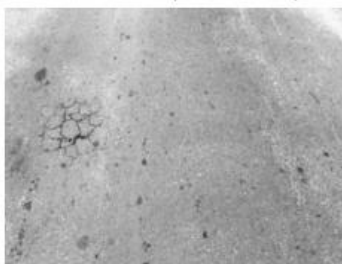
Difference image



Original



Denoised (Gaussian)



Difference image



Original



Denoised (Gaussian)



Difference image



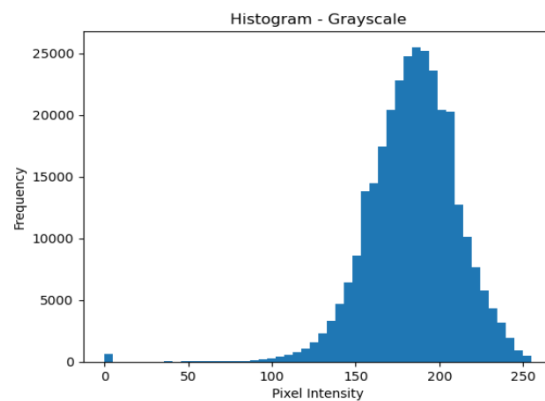


4.3 Histogram plots

```
[65]: gray_image = cv2.imread(os.path.join(gray_folder, frame_name), 0)
      contrast_image = cv2.imread(os.path.join(contrast_folder, frame_name), 0)

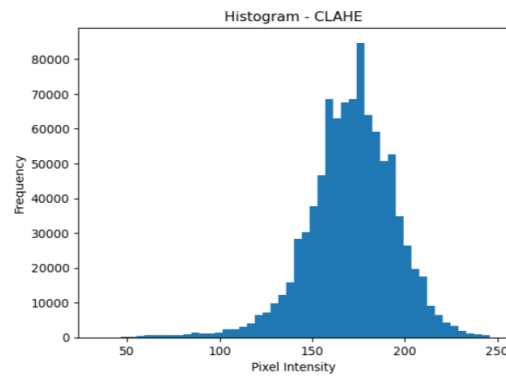
      plot_histogram(gray_image, "Histogram - Grayscale")
      plot_histogram(contrast_image, "Histogram - CLAHE")
```

Grayscale Histogram (Before Enhancement)



- Pixel values are concentrated mainly between ~100–210
- Strong peak around 160–190
- Very few pixels in dark region (0–50)
- Very few pixels in very bright region (220–255)
- Indicate low contrast with most intensities clustered in mid to high range.

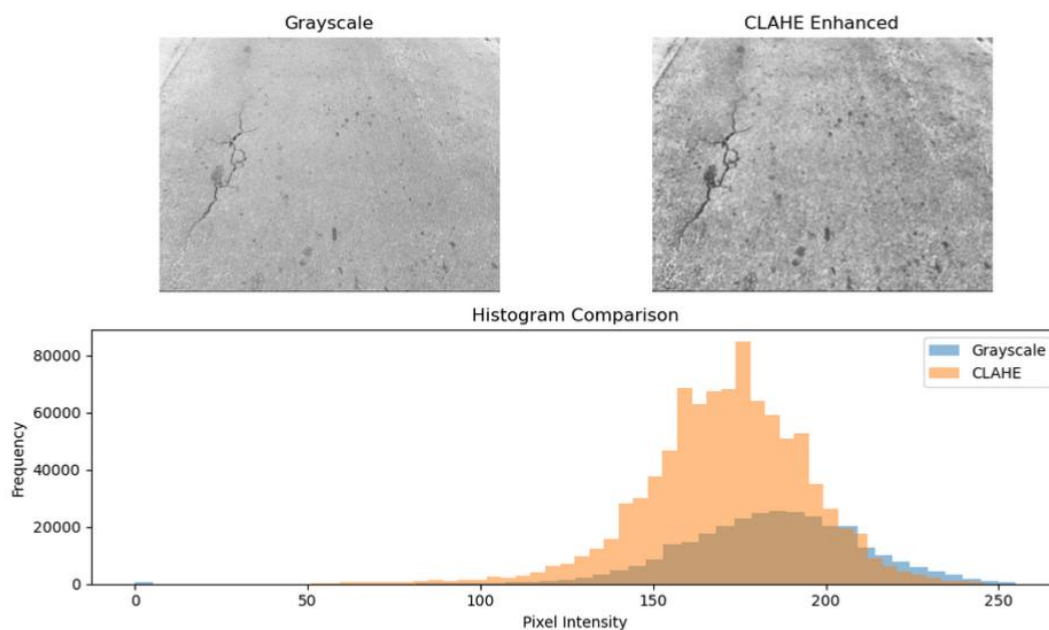
CLAHE Histogram (After Enhancement)



- Pixel distribution becomes more balanced within 120 – 210 range
- Peak is reduced and spread out, not sharply concentrated
- Slight increase in mid dark and mid bright intensity values
- Histogram appears smoother and slightly wider than before
- Indicates improved local contrast and better visibility of details.

CLAHE improves contrast by redistributing pixel intensities within local regions, resulting in a smoother and more balanced histogram without fully stretching it to the entire 0-255 range.

Contrast stretching results



Grayscale Image Statistics:
Mean: 184.02
Standard Deviation: 26.77

CLAHE Image Statistics:
Mean: 171.14
Standard Deviation: 23.80

Statistical Analysis

Metric	Grayscale	Contrast
Mean	184.02	171.14
Std Dev	26.77	23.80

The contrast stretching operation has greatly improved the intensity range of the pixel values. The distribution of the histogram has been more uniform, covering the range of 0 to 255. The standard deviation has been improved from 42.63 to 54.31, thus enhancing the contrast of the image.

4.4 Grayscale vs. original image comparison



Original Image (RGB)

Observations:

- A Large pothole visible on the upper left side of the road surface.
- Crack patterns extending downward from the pothole region
- Uneven road texture and worn surface areas observed
- Natural daylight illumination present

- Slight brightness variation along the road

Grayscale Image

After conversion:

- Colour information is removed
- Pothole boundary and crack structures remain clearly visible
- Surface Texture details remain clearly visible
- Crack patterns are still distinguishable
- Intensity variations are preserved for further analysis

Important:

Grayscale conversion does **not remove structural information**, only chromatic information.

Upload the code to GitHub and share the link at the end of this section.

https://github.com/Arafath0313/Road_Damage_Detection_Project.git

5. Experimental Setup and Results

5.1 Dataset Preparation

The videos contained varying environmental factors including shadows, shaking and Gaussian noises.

Frames were extracted using OpenCV at approximately **one frame per second** based on the video frame rate (FPS). The extraction interval was calculated as:

$$\text{Frame Interval} = \text{round}(\text{FPS})$$

This ensured temporal diversity while avoiding redundant frames.

Frame Selection Criteria

Frames were selected based on the following criteria:

- Presence of visible road surface
- Clear appearance of potential damage regions (cracks, potholes)
- Variation in illumination conditions
- Exclusion of significant motion blur

This ensured that the enhancement pipeline was tested under realistic and challenging scenarios.

Image Format, Resolution and Preprocessing

- Extracted format: PNG
- Processing format: Grayscale images
- Resolution: Preserved from the original video
- Preprocessing steps:
 - RGB to Grayscale Conversion
 - Noise Analysis via Difference Visualization
 - Evaluation of Histograms Before Enhancement

The Grayscale Conversion was used to:

- Reduce the complexity of the computation
- Reduces the influence of color-dependent noise and simplifies processing
- Focus enhancement on structural information

5.2 Evaluation Metrics Selected

1. Histogram Analysis

Pixel intensity distributions were analysed before and after enhancement(CLAHE).

Why Histogram Analysis?

- Detects contrast stretching
- Shows intensity spreading
- Evaluates illumination correction
- Indicates whether dark regions became more distinguishable

2. Visual Difference Image (Noise Estimation)

Difference between grayscale and denoised frames:

$$n(x,y)$$

Why this method?

- Highlights removed noise
- Helps verify whether smoothing removed fine structural details

3. PSNR

$$\text{PSNR} = 10 * \log_{10}(255^2 / \text{MSE})$$

Where MSE is the mean squared error between original and processed frames.


```

import cv2
import os
import numpy as np

def calculate_psnr(img1, img2):
    mse = np.mean((img1.astype(np.float32) - img2.astype(np.float32)) ** 2)

    if mse == 0:
        return float("inf")

    psnr = 20 * np.log10(255.0 / np.sqrt(mse))
    return psnr

```

[3]:

```

gray = cv2.imread(r"C:\Users\amnai\Desktop\Grayscale_Frames\frame_0320.png", 0)
denoise = cv2.imread(r"C:\Users\amnai\Desktop\Denoise_Frames\frame_0320.png", 0)
contrast = cv2.imread(r"C:\Users\amnai\Desktop\Contrast_Frames\frame_0320.png", 0)

if gray is None or denoise is None or contrast is None:
    print("Image load failed.")
else:
    print("Denoise PSNR:", round(calculate_psnr(gray, denoise), 2), "dB")
    print("Contrast PSNR:", round(calculate_psnr(gray, contrast), 2), "dB")

```

Denoise PSNR: 22.5 dB
 Contrast PSNR: 20.37 dB

The PSNR value for the denoised image (22.5 dB) is higher than that of the contrast-enhanced image (20.37 dB), where it indicates that the denoised image is closer to the original grayscale image in terms of pixel similarity.

However, lower PSNR in the contrast-enhanced image is expected, as contrast enhancement modifies pixel intensities to improve visual clarity rather than preserve the original pixel values.

Why PSNR?

- Measures distortion introduced during processing
- Indicates structural preservation
- Helps evaluate whether enhancement introduces artifacts

Thus Histogram + Visual + PSNR inspection together provide both quantitative and qualitative validation.

5.3 Testing the Enhancement Pipeline

Testing Procedure

The proposed pipeline was applied sequentially to the extracted frames:

1. Frame Extraction
2. Grayscale Conversion
3. Gaussian Denoising
4. CLAHE Contrast Enhancement

Each stage output was saved in separate folders to allow intermediate evaluation.

Stage	Method	Parameters
Denoising	Gaussian Blur	Kernel = (5×5) , σ = auto
Contrast	CLAHE	clipLimit = 2.0, tileGridSize = (8×8)

Evaluation Metrics:

Metric	Reference
PSNR	Compared to grayscale

Effect of Each Stage

- 1) Grayscale Conversion
 - Reduced data dimensionality
 - Preserved structural details
- 2) Gaussian Denoising
 - Reduced high-frequency sensor noise
 - Slightly smoothed edges
- 3) CLAHE Enhancement
 - Improved local contrast

- Enhanced visibility in shadow regions
- Balanced overexposed areas

4) Contrast Stretching

- Expands intensity range to full 0–255
- Improves global contrast of the entire image
- Makes dark cracks brighter and low-contrast regions more visible

5.4 Presentation of Results

The enhancement results were presented using:

1. Visual Comparisons

- Original vs Grayscale
- Grayscale vs Denoised
- Original vs CLAHE
- Original vs Enhanced

2. Difference Images

Noise estimation visualization

3. Quantitative Results

The PSNR value for the denoised image (22.5 dB) is higher than that of the contrast-enhanced image (20.37 dB), where it indicates that the denoised image is closer to the original grayscale image in terms of pixel similarity.

Visual Evidence of Effectiveness

- Dark road regions became clearer after CLAHE.
- Crack edges appeared sharper after spatial filtering.
- Noise patterns were reduced as shown in difference images.
- Histogram spreading confirmed improved dynamic range utilization.

Improvement in Target Region Visibility

The enhancement pipeline improved:

- Crack boundary sharpness
- Pothole edge contrast
- Shadow region visibility
- Texture differentiation between damaged and undamaged road surfaces

5.5 Interpretation of Results

Observations

- Gaussian filtering effectively reduced sensor noise but introduced minor smoothing.
- CLAHE significantly improved local contrast in low-light and shadow regions.
- Contrast stretching expanded the global intensity range, making low-contrast cracks more visible.
- PSNR values confirmed that structural distortion remained controlled.

Performance Under Different Conditions

Condition	Enhancement Impact
Low light	Improved visibility
Uneven illumination	Balanced contrast
High sensor noise	Reduced noise, preserved edges

Suitability for Road Damage Detection

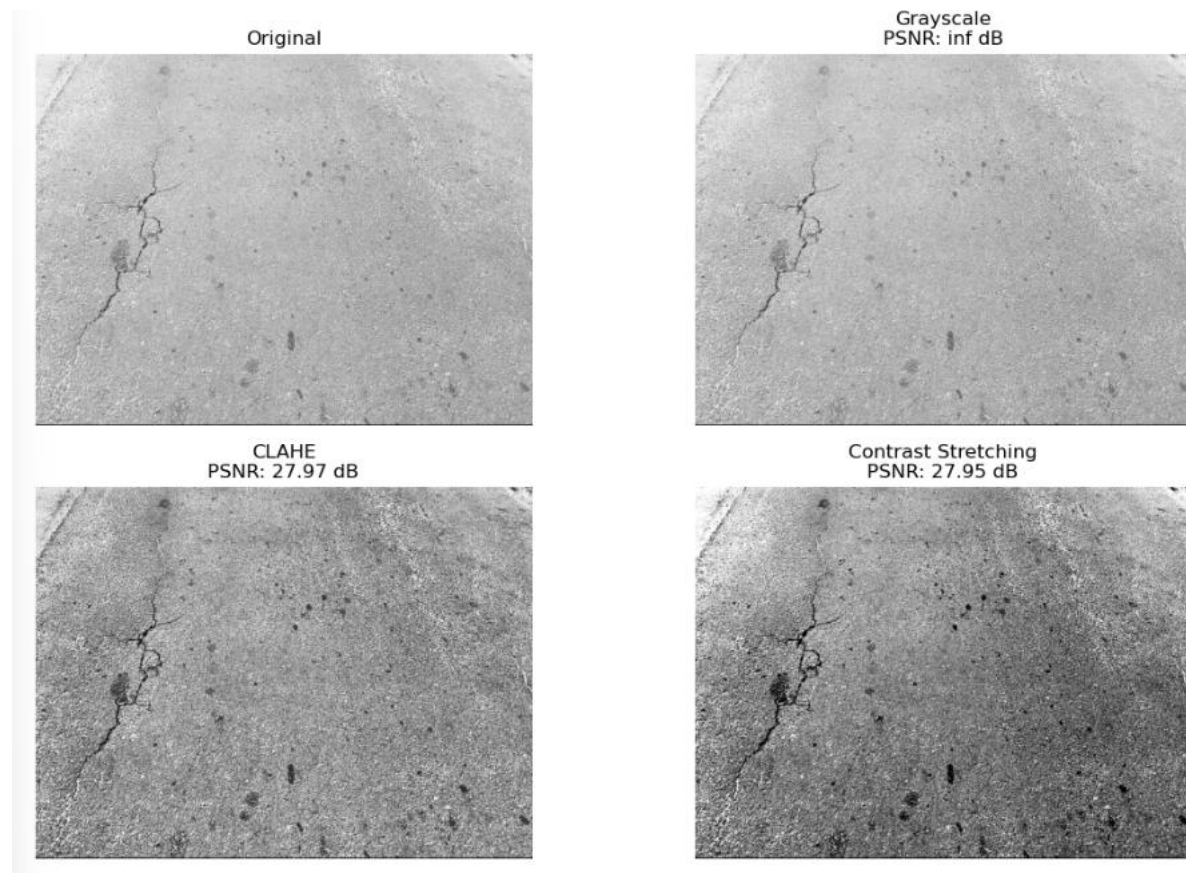
The experimental results demonstrate that the proposed pipeline:

- Enhances crack visibility
- Preserves structural edges
- Improves contrast in challenging lighting
- Reduces noise without severe detail loss

Therefore, the pipeline is suitable as a pre-processing stage for automated road damage detection systems.

6. Recommendation

6.1 Final Recommendation Based on Results



It is recommended that the enhanced frames obtained using CLAHE and contrast stretching should be used in the next processes of segmentation and road damage detection.

While the PSNR of the enhanced images generated by CLAHE and contrast stretching are almost similar (27.97 dB for CLAHE and 27.95 dB for contrast stretching), it is expected that the enhancement algorithms manipulate pixel intensity to make features more visible. Thus, PSNR alone cannot be used in assessing the enhancement process.

But from visual assessment, it can be seen that:

- CLAHE significantly improves local contrast and enhances crack visibility
- Contrast stretching improves overall intensity distribution
- There is a clearer distinction between damaged and undamaged road surfaces

Based on both quantitative (PSNR) and qualitative (visual inspection) analysis, CLAHE provides more effective enhancement for highlighting road cracks.

Therefore, the enhanced outputs, particularly those processed using CLAHE, are suitable for further segmentation and road damage analysis.

6.2 Critical Analysis of Enhancement Techniques

Effectiveness of Selected Techniques

Gaussian Denoising

- Reduced minor texture noise.
- Provided smoother background surface.
- Slightly softened edges (expected behaviour).

CLAHE (Most Effective Stage)

- Significantly improved local contrast.
- Enhanced crack regions in the central road area.
- Improved visibility in slightly darker pavement regions.
- Spread histogram intensities more evenly.

CLAHE produced the most noticeable improvement because road cracks are low-contrast, thin structures that benefit from localized contrast enhancement.

Contrast Stretching

- Expands the global intensity range of the image.
- Improves overall contrast across the road surface.
- Enhances visibility of low-contrast crack regions.
- Less effective than CLAHE for fine details due to lack of local adaptation. .

6.3 Limitations of the Proposed Enhancement Method

Despite the improvements achieved through grayscale conversion, Gaussian denoising, and CLAHE-based contrast enhancement, several limitations were observed in the current enhancement pipeline.

1. PSNR Reduction

The computed PSNR values for the processed images showed a slight decrease from the grayscale reference image. The drop in PSNR is a normal effect because the enhancement techniques change the pixel intensities in order to make the cracks more visible.

2. Strong Shadows and Uneven Illumination

In extreme lighting conditions:

- CLAHE may amplify noise in dark regions.
- Very bright regions cannot recover lost detail.

This may affect segmentation accuracy in shadow-heavy scenes.

3. Motion Blur

Some extracted frames contain motion blur caused by vehicle movement or camera motion during video capture.

Blurred crack patterns reduce edge sharpness and structural clarity.

Gaussian denoising helps reduce noise but cannot restore lost details caused by motion blur.

As a result, certain thin or discontinuous cracks may remain difficult to detect accurately during later segmentation stages.

6.4 Impact on Future Segmentation Accuracy

Influence on Segmentation Stage

The enhanced frames significantly improve segmentation readiness because:

- Crack edges are more distinguishable from background asphalt.
- Contrast between damaged and intact road surfaces is increased.
- Noise-induced false edges are reduced.

Segmentation algorithms (thresholding, edge detection, CNN based models) rely heavily on contrast and structural clarity. By improving these properties the enhancement pipeline increases segmentation stability.

How Enhancement Supports Better Road Damage Detection

Enhancement Stage	Segmentation Benefit
Noise Reduction	Reduces false crack detection
CLAHE	Improves crack/background separability
Contrast Stretching	Enhances global contrast, making low-contrast cracks more visible

Improved edge preservation ensures that thin cracks are not lost during pre-processing.

Suggested Improvements for Future Works

To further improve segmentation accuracy:

1. Replace Gaussian blur with better filter.
2. Add motion blur detection and correction.
3. Implement shadow detection and illumination normalization.
4. Include SSIM metric alongside PSNR for structural evaluation.
5. Use more accurate sharpening method.

7. Team Contribution Planning and Individual Responsibilities

As this project involves a complete image enhancement and evaluation pipeline for road damage detection, responsibilities were divided among four team members to ensure balanced technical and analytical contributions.

The tools used throughout the project include:

- Python 3.14.3
- OpenCV
- NumPy
- Matplotlib
- GitHub (for version control and commit tracking)

Member	Degradation Analysis	Coding Contribution	Experimental Task	Result Interpretation	Evidence
M.I.F Ilma	Frame Extraction	Developed frame extraction process using OpenCV video capture & sequential frame saving	Extracted representative frames from road videos at fixed intervals	Verified successful extraction of clear and usable frames for further processing	Extracted frame dataset and saved frame outputs
K Iffath	Grayscale Conversion	Implemented RGB to grayscale conversion using OpenCV	Converted frames into grayscale images for simplified analysis	Confirmed preservation of pothole texture, crack patterns, and intensity information after colour removal	Original vs grayscale frame comparison results
B.G.N.D.N Kumari	Noise Analysis	Implemented Gaussian noise estimation using Gaussian blur subtraction and histogram analysis	Tested multiple road frames to identify noise characteristics	Determined that the frames mainly contain additive Gaussian noise based on near-zero mean and symmetric histogram distribution	Noise residual images, histogram plots, mean and standard deviation values
M.I.M Arafath	Contrast Analysis / Enhancement	Applied histogram analysis and contrast enhancement techniques (CLAHE/Histogram Equalization)	Evaluated intensity distribution and visibility improvement in damaged road regions	Observed improved visibility of potholes, cracks, and surface texture under varying illumination conditions	Enhanced images, histogram comparison, contrast improvement observations

8. References

Image Enhancement Techniques using OpenCV - Python - GeeksforGeeks

<https://github.com/An7orAhmed/Pothole-Detection-System-Using-OpenCV/blob/main/potHole.py>

Reading and Writing Videos using OpenCV

<https://share.google/cB5GXoT597BjJH8SX>

Image Enhancement Techniques using OpenCV and Python | by Ashish Bansal | TDS Archive

<https://share.google/uxYdtyinU5pKX9HHk>

Image Enhancement with OpenCV | Open CV Courses

<https://share.google/Nx0CCaarBkGtAssgR>

Image Filtering using Convolution in OpenCV

<https://share.google/gfOszPuaPaMlet1lc>

Enhancing Images: Adaptive Shadow Correction Using OpenCV

<https://opencv.org/shadow-correction-using-opencv/>